

One Evidence Standard Now. One Exact Benchmark Next.

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The proposal. Require every State to publish a reproducible geographic benchmark, its final congressional plan, every changed block, alternatives, effects, public comments, and reasons.

The North Star. Replace the heuristic benchmark with an exact, uniquely selected plan backed by a mathematical proof certificate.

What We Can Defend Today

BISECT can produce and verify a partisan-input-excluded geographic benchmark. It records code, inputs, parameters, assignments, analysis, and reports.

That claim is useful but limited:

- it proves execution, not fairness;
- it does not establish Voting Rights Act compliance;
- it does not preserve every community;
- it does not prove a global optimum; and
- it does not replace a commission, court, or public process.

The model federal bill therefore treats the benchmark as mandatory *evidence*, not the mandatory final map.

Why This Is Still A Major Reform

Today, a final map can hide thousands of choices. Under the proposed standard, the public receives:

1. one precommitted benchmark;
2. one legally adopted final plan;
3. a block-by-block diff;
4. legal authority for every change;
5. a lower-departure lawful alternative for discretionary changes;
6. public and independent review; and
7. a permanent challenge record.

The benchmark cannot be used as a VRA safe harbor. Federally required remedies are judged against full legal compliance, not proximity to a race-excluded map.

What The Real Evidence Shows

The evidence program has become more credible because it reports failures.

- Governed NRS v0.3 packages assign all 27,398,654 Census blocks across the 2000, 2010, and 2020 cycles: 50 States, 435 districts, and 385 recursive nodes per cycle.
- Independent package verification found zero omitted or duplicate units, zero disconnected districts, and zero population-tolerance failures.
- Only 13 of 1,155 recursive nodes attain an arithmetic exact-population floor. Exact weighted-boundary and canonical proof coverage remains zero.
- A precommitted bounded comparison retained all 40 METIS rows. METIS matched the complete exact primary objective in 30 and the canonical assignment in 20; the result does not extrapolate to State-scale optimality.
- A separate Python implementation verifies the three-cycle structural comparison: 120 node signatures are common to every cycle and 18 States retain the exact recursive topology in all three.
- A precommitted county/tract audit publishes and independently verifies 231,765 geography rows. County splits are 1,812 of 3,140 in 2000, 1,819 of 3,142 in 2010, and 1,823 of 3,142 in 2020; tract splits are 17,268 of 65,255, 18,800 of 72,878, and 20,288 of 84,208. These are within-vintage descriptive counts, not a trend or legal-compliance claim.
- A separately governed 2020 bakeoff compares NRS v0.3 with official CD118 on 7,889,194 common land-containing blocks. After State-level label matching, 4,194,107 block assignments match and 3,695,087 differ (53.162680% agreement); NRS splits 1,808 counties and 19,789 tracts, versus 404 and 4,720 in the comparator block projection.
- The same common-block universe supplies four geometry measures. Both State- and district-weighted summaries place CD118 above NRS on Polsby–Popper, exact Reock, and convex-hull ratio and below NRS on

Schwartzberg. This is not original enacted linework and does not identify a fairer, better, or legally preferred plan.

- Raw tree hashes reproduce for 149 of 150 State-cycle packages. Maryland 2010 repeats the exact assignment but not one diagnostic move counter, so the published audit uses its assignment hash and committed substantive node snapshot and discloses the transport-hash exception.
- A fresh-clone second-laptop kit can verify committed artifacts or replay all national assignments. An actual external v0.3 record is still pending.
- Earlier real Rust and GerryChain ensembles remain archived for Rhode Island, Iowa, and North Carolina. Their limited scope and convergence failures remain part of the record.
- Rhode Island supports a stable low cut-fraction result.
- Iowa and North Carolina lack enough effective samples for extreme-tail claims.
- Rust and GerryChain distributions differ materially in Iowa and North Carolina.
- Wisconsin was excluded because a benchmark district was disconnected.

These are not marketing defects. They show that the evidence system can reject its own preferred story.

The Huntington–Hill North Star

Huntington–Hill ended operational dispute after Congress selected the method. Redistricting is harder because it creates geographic regions rather than allocating seats among fixed States.

The closest equivalent is an **Exact Canonical Benchmark**:

1. Congress enacts the feasible-set axioms.
2. An exact solver minimizes maximum population deviation.
3. It then minimizes total population deviation.
4. It then minimizes geographic boundary cost.
5. A canonical GEOID assignment vector breaks all remaining ties.
6. A proof certificate establishes that no lexicographically better assignment exists under the identified enacted instance, model, and profile.

Given the enacted axioms and census inputs, the benchmark would be unique and mathematically certified. Debate remains over the axioms and lawful modifications—not over what the mathematics produced.

Gold-standard statement:

“For this enacted instance, model, and profile, this is the unique assignment selected by the enacted lexicographic objective. The attached certificate proves feasibility, model-relative optimality, and canonical tie-breaking.”

Why We Are Not There Yet

Exact connected redistricting is computationally difficult. NRS v0.3 is an operational deterministic procedure whose national assignments and tolerance claims verify; it is not an exact optimizer. Bounded exact split and recursive-tree certificates, hostile verifier corpora, independent verifiers, OPB proof requests, and national block RCTX inputs are implemented. They are foundations, not a complete national exact solver.

The North Star requires:

- scalable block-level exact formulations;
- compact proof-loggable connectivity constraints;
- lower bounds matching the incumbent objective;
- canonical tie-optimality proofs;
- RoundingSat proof generation and independent VeriPB verification; and
- demonstrated exact-proof performance from small States to California and Texas.

Adoption Path

Actor	Near-term use
Congress	Enact the benchmark, disclosure, governance, and exact-standard research program.
Commissions	Use the benchmark as a public comparison and explain every departure.
Courts	Use packages and audits as evidence; retain legal judgment.
States	Precommit criteria, publish alternatives, and fund community participation.
Civic groups	Submit communities and plans; challenge data, procedure, and substantive harms.
Researchers	Reproduce the evidence and attack the exact-certificate design.

The Ask

Do not wait for exact optimization before adopting transparency. Adopt the evidence standard now.

Do not stop at a heuristic benchmark. Fund and test the Exact Canonical Benchmark until it can supply Huntington–Hill-like operational finality.

Claim Boundary

This brief describes an internal candidate standard. It is not enacted law, external peer review, independent physical-laptop replication, or proof that a BISECT plan is fair. Block-level operational execution is complete for the three governed national cycles; exact national boundary/canonical certification and the full neutral evaluation schedule are not. The exact benchmark remains a research and engineering goal.